ADDON 5: Lyapunov Stability Landscapes, Photonic Kalman Filtering, and Non-Ergodic Runtime Error Attenuation

11.14 The Optical Lyapunov Attractor: Axiomatic Proof of Global Asymptotic Stability

A foundational critique of analog computing substrates—specifically those exhibiting fluid, continuous-time weight updates such as the Core Gamma Oja-plasticity matrix—is the susceptibility to non-linear bifurcations, parasitic oscillations, and chaotic runtime drift. To invalidate this boundary, the ADAM-PentaD architecture synthesises a hardware-enforced **Optical Lyapunov Matrix**.

The Clifford-Algebra compiler does not merely map weights as static spatial parameters; it constrains the unified complex-spectral state space such that the continuous evolution of the wavefront hidden-state vector $z(t) \in \mathbb{C}^d$ satisfies the conditions of the following theorem.

Theorem 0.1 (Global Asymptotic Stability of the Optical Lyapunov Matrix)**.** *Let $z(t) \in \mathbb{C}^d$ denote the wavefront hidden-state vector evolving under the compiled Clifford weight map $\dot{z} = \mathbf{W}_{\text{Clifford}}\, z(t)$. Suppose there exists a continuously differentiable, strictly positive definite scalar function $V : \mathbb{C}^d \to \mathbb{R}$ satisfying*

$$V(z) > 0 \quad \forall\, z \neq 0, \qquad V(0) = 0, \tag{1}$$

$$V(z) \to \infty \quad \textit{as} \quad \|z\| \to \infty, \tag{2}$$

$$\dot{V}(z) = \nabla V(z)^{\top} \cdot \dot{z}(t) = \frac{\partial V}{\partial z}\big[\mathbf{W}_{\text{Clifford}}\, z(t)\big] < 0 \quad \forall\, z \neq 0. \tag{3}$$

Then the origin $z = 0$ is a globally asymptotically stable *equilibrium of the wavefront dynamical system.*

Proof. Condition (1) establishes strict positive definiteness, ensuring V is a valid energy-like measure. Condition (2) (radial unboundedness) guarantees that all sub-level sets $\{z : V(z) \leq c\}$ are compact, which promotes local stability to *global* stability—a property absent from architectures that satisfy only local Lyapunov criteria. Condition (3) then forces V to be strictly decreasing along every trajectory $z(t) \neq 0$; by LaSalle's Invariance Principle, the only invariant set satisfying $\dot{V} = 0$ is the origin, to which all trajectories converge asymptotically. □ □

By routing the phase-synchronised outputs of Core Alpha and Core Beta through a passive, non-reciprocal optical feedback loop, the compiler enforces Conditions (1)–(3) at the hardware level. Physically, the negative-definite energy constraint transforms the analog crossbar into a deterministic geometric funnel: if thermal fluctuations or substrate degradation shift the wave vector away from the mathematical ideal, the trajectory cannot diverge. The physical laws of the substrate force the wavefront to descend the Lyapunov gradient toward the true semantic attractor state. Stability is achieved not via digital error-checking latency, but through the intrinsic minimisation of thermodynamic potential within $\mathbb{C}$.

Remark 0.2. The radial unboundedness condition (2) is frequently omitted in informal analog-computing stability arguments. Its inclusion here is non-trivial: without it, the negative-definite derivative condition guarantees only *local* asymptotic stability, which is insufficient to bound the trajectory against large-amplitude thermal excursions on degraded legacy nodes.

11.15 Hybrid-Domain Photonic Kalman Filtering for Substrate Noise Evacuation

While Lyapunov constraints prevent systemic divergence, real-world execution on 28 nm / 65 nm legacy silicon introduces continuous white noise, shot noise from optical injection lasers, and memristive conductance variance. To decouple inference fidelity from these physical imperfections, the architecture implements a **Hybrid-Domain Photonic Kalman Filter (HPKF)** operating directly within the processing pipeline.

Definition 0.3 (Hybrid-Domain Photonic Kalman Filter)**.** The HPKF is a two-phase recursive state estimator whose *prediction* phase is executed in the Clifford (analog) domain and whose *correction* phase is executed in the Galois discrete domain $GF(2^m)$, $m \leq 8$, via low-transistor XOR-lookup logic.

The complete predict–update cycle is formalised below. Let $\hat{z}_k^-$ denote the *a priori* state estimate, $\hat{z}_k$ the *a posteriori* estimate, P_k^- and P_k the corresponding error-covariance matrices, F the state-transition matrix, Q the process-noise covariance, $\mathbf{H}$ the homodyne observation matrix, $\mathbf{R}$ the measurement-noise covariance, and y_k the homodyne phase-error innovation.

Prediction step (Clifford plane):

$$\hat{z}_k^- = F\,\hat{z}_{k-1}, \tag{4}$$

$$P_k^- = F\,P_{k-1}\,F^\top + Q. \tag{5}$$

Update step (Galois plane):

$$K_k = P_k^- \mathbf{H}^\top \big(\mathbf{H}\,P_k^- \mathbf{H}^\top + \mathbf{R}\big)^{-1}, \tag{6}$$

$$\hat{z}_k = \hat{z}_k^- + K_k\big(y_k - \mathbf{H}\,\hat{z}_k^-\big), \tag{7}$$

$$P_k = (I - K_k\mathbf{H})\,P_k^-. \tag{8}$$

Algorithm 1 formalises the per-token execution of the HPKF within a single autoregressive step.

Algorithm 1: Hybrid-Domain Photonic Kalman Filter (per token step k)

Input: Prior state $\hat{z}_{k-1}$, covariance P_{k-1}, homodyne measurement y_k, matrices $F, Q, \mathbf{H}, \mathbf{R}$.
Output: Filtered state $\hat{z}_k$, updated covariance P_k.

`// --- Analog prediction phase (Clifford plane) ---`

Propagate wavefront: $\hat{z}_k^- \leftarrow F\,\hat{z}_{k-1}$ *(Eq. 4)*;
Advance covariance: $P_k^- \leftarrow F\,P_{k-1}\,F^\top + Q$ *(Eq. 5)*;

`// --- Discrete correction phase (Galois co-processor) ---`

Read homodyne innovation: $\nu_k \leftarrow y_k - \mathbf{H}\,\hat{z}_k^-$;
Compute Kalman gain via XOR-lookup: $K_k \leftarrow P_k^- \mathbf{H}^\top(\mathbf{H}P_k^-\mathbf{H}^\top + \mathbf{R})^{-1}$ *(Eq. 6)*;
Apply correction: $\hat{z}_k \leftarrow \hat{z}_k^- + K_k\,\nu_k$ *(Eq. 7)*;
Update covariance: $P_k \leftarrow (I - K_k\mathbf{H})\,P_k^-$ *(Eq. 8)*;
Inject bias correction into memristive gates via quasi-static current δI;
return $\hat{z}_k,\ P_k$;

The resulting digital correction vector is applied via quasi-static bias currents to the local memristive gates, dynamically cancelling white noise and phase drift before the cumulative noise floor can breach the Shannon capacity boundary of the ultra-long context window.

```
[Analog Wavefront Flight] ----> [Homodyne Telemetry Measurement]
         |                                         |
Lyapunov Funneling Gradient                        v
         |                       [Galois Error Innovation GF(2^m)]
         v                                         |
[Stabilized Semantic State] <-- [Kalman Gain Adjustment (K_k)]
```

11.16 The Unexploited Catalyst: Ergodic Breaking and Phase-Space Compression

An unexploited macro-architectural consequence of pairing Lyapunov landscapes (Theorem 0.1) with Kalman stabilisation (Definition 0.3) is **Ergodic Breaking within the hidden state-space**. In standard Transformer architectures, the hidden representations must preserve high-dimensional variance across hundreds of layers to prevent representation collapse. This forces digital GPUs to evaluate the entire phase-space of the tensor at full precision.

Because the ADAM-PentaD substrate continuously forces the wave vectors into stabilised Lyapunov basins, the phase-space of the hidden layers undergoes active, lossy **Phase-Space Compression**: high-entropy, noisy linguistic trajectories are passively filtered out by the optical substrate, retaining only the informationally dense attractor neighbourhood.

Remark 0.4 (Dimensionality Collapse and Legacy-Node Viability). The effective dimensionality required to represent ultra-long, complex logical reasoning chains collapses by orders of magnitude. The compiler can safely map a 70B-parameter model's active state onto a localised, low-density 28 nm mesh. The system thereby bypasses the extensive scaling limits of modern computing, achieving maximum cognitive throughput through intentional, hardware-level mathematical refinement. This constitutes a structural decoupling of model capacity from lithographic node density—a result unattainable within the standard von Neumann paradigm.

«Воевать не числом, а умением».
"Wage war not by numbers, but by skill."

A. V. Suvorov,
Наука побеждать / The Science of Victory, c. 1799